Congruences — Solutions

Problem 1. Find all positive integers a, b, c such that

$$a \equiv b \pmod{c}, \quad b \equiv c \pmod{a}, \quad c \equiv a \pmod{b}.$$

Solution. WLOG assume $a \geqslant b, c$. Then $b \equiv c \pmod{a}$ implies $b = c$. The other conditions become $a \equiv 0 \pmod{b}$. Thus, a can be any multiple of b. The general solutions are

$$(a, b, c) = (km, k, k), (k, km, k), (k, k, km)$$

where k, m are positive integers. □

Problem 2. Let $n \geqslant 5$ be a squarefree integer. Prove that there exists an integer m such that

$$m^4(m^4 - 5) \equiv 36 \pmod{n}.$$

Solution. Note that

$$m^4(m^4-5) - 36 = (m^4-9)(m^4+4) = (m^2-3)(m^2+3)\big((m+1)^2+1\big)\big((m-1)^2+1\big).$$

Let $n = p_1 p_2 \cdots p_s$ where each p_j is a prime. For each p_j, if $p_j \equiv 1 \pmod{4}$, there exists x_j such that $(x_j+1)^2 \equiv -1 \pmod{p_j}$ since -1 is a quadratic residue modulo p_j. If $p_j \equiv -1 \pmod{4}$, then $\left(\dfrac{-1}{p_j}\right) = -1$. Thus, $\left(\dfrac{3}{p_j}\right) = 1$ or $\left(\dfrac{-3}{p_j}\right) = 1$. This means we have $x_j^2 \equiv 3 \pmod{p_j}$ or $x_j^2 \equiv -3 \pmod{p_j}$ for some x_j. It follows that $x_j^4(x_j^4 - 5) \equiv 36 \pmod{p_j}$ in any case. By the Chinese remainder theorem, there exists m such that $m \equiv x_j \pmod{p_j}$ for all j. This m is the one we want. □

Problem 3. (Balkan MO Shortlist 2021 N3) Let n be a positive integer. Determine, in terms of n, the greatest integer which divides every number of the form $p + 1$, where $p \equiv 2 \pmod{3}$ is a prime number which does not divide n.

Solution. Let M be the desired integer.

If n is odd, we can take $p = 2$. Then $M \mid 2 + 1 = 3$, and so $M \leqslant 3$. Conversely, 3 divides every $p + 1$ as $p + 1 \equiv 0 \pmod{3}$. This shows $M = 3$.

If n is even, p must be odd. Therefore, $p \equiv 5 \pmod{6}$, and so 6 always divides $p + 1$. We show that $M \leqslant 6$. By Dirichlet's theorem, there exists a prime $p > n$ such that $p \equiv 29 \pmod{36}$. Note that $p \equiv 2 \pmod{3}$. For any such p, it is clear that $p \nmid n$. Since $p + 1 \equiv 30 \pmod{36}$, we have $4 \nmid M$ and $9 \nmid M$. Similarly, for any prime $q > 3$, there exists a prime $p > n$ such that $p \equiv 2 \pmod{3}$ and $p \equiv 1 \pmod{q}$. For this p, we have $p \nmid n$ and $p+1 \equiv 2 \not\equiv 0 \pmod{q}$. Therefore, $q \nmid M$. Combining these, we must have $M \leqslant 6$, and hence $M = 6$. □

Problem 4. (Indonesia MO 2022 Problem 5) Let $N \geqslant 2$ be a positive integer. Given a sequence of natural numbers $a_1, a_2, \ldots, a_{N+1}$ such that for all integers $1 \leqslant i \leqslant j \leqslant N+1$,

$$a_i a_{i+1} a_{i+2} \cdots a_j \not\equiv 1 \pmod{N}.$$

Prove that there exists a positive integer $k \leqslant N+1$ such that $\gcd(a_k, N) \neq 1$.

Solution. Consider the $N+1$ integers

$$a_1, a_1 a_2, a_1 a_2 a_3, \ldots, a_1 a_2 a_3 \cdots a_{N+1}.$$

By the pigeonhole principle, $a_1 \cdots a_{i-1} \equiv a_1 \cdots a_j \pmod{N}$ for some $i \leqslant j$. If $a_1, a_2, \ldots, a_{i-1}$ are relative prime to N, then this implies $1 \equiv a_i a_{i+1} \cdots a_j \pmod{N}$, contradiction. Therefore, we must have $\gcd(a_k, N) \neq 1$ for some k. $\square$

Problem 5. (Romania 1996) Find all prime numbers p, q for which the congruence

$$\alpha^{3pq} \equiv \alpha \pmod{3pq}$$

holds for all integers α.

Solution. The congruence implies $\alpha^{3pq} \equiv \alpha \pmod{p}$. Choosing α to be a primitive root modulo p, this implies $\alpha^{3pq-1} \equiv 1 \pmod{p}$, and hence $p-1 \mid 3pq-1$. This gives $p-1 \mid 3q-1$. By symmetry, we have $q-1 \mid 3p-1$. WLOG assume $p \geqslant q$.

- If $q = 2$, then $p-1 \mid 5$. This gives $p = 2$. But then $2^{12} \equiv 2 \pmod{12}$ does not hold.
- If $q = 3$, then $p-1 \mid 8$ and $2 \mid 3p-1$. This gives $p = 3, 5$. But then both $3^{27} \equiv 3 \pmod{27}$ and $3^{45} \equiv 3 \pmod{45}$ do not hold.
- If $q \geqslant 5$, then $4(p-1) \geqslant 4q-4 > 3q-1$. Therefore, from $p-1 \mid 3q-1$, we must have $3q-1 = p-1, 2(p-1), 3(p-1)$.
  - If $3q-1 = p-1$, then $p = 3q$ is not a prime.
  - If $3q-1 = 2(p-1)$, then $q-1 \mid 3p-1$ implies $q-1 \mid 2(3p-1) = 6(p-1)+4 = 9q+1$. This gives $q-1 \mid 10$, and hence $q = 11$ and $p = 17$. Conversely, we now check that $\alpha^{561} \equiv \alpha \pmod{561}$ for all integers α. Indeed, we have $2 \mid 560$, $10 \mid 560$ and $16 \mid 560$. Thus, $\alpha^{561} \equiv \alpha \pmod{r}$ for $r = 3, 11, 17$ by the Fermat little theorem. The claim follows from the Chinese remainder theorem.
  - If $3q-1 = 3(p-1)$, then q is not an integer.

Therefore, the only solutions are $(p, q) = (11, 17), (17, 11)$. $\square$

Problem 6. Let p be a prime number. Prove that

$$\sum_{j=0}^{p} \binom{p}{j}\binom{p+j}{j} \equiv 2^p + 1 \pmod{p^2}.$$

Solution. For $1 \leqslant j \leqslant p-1$, we have $p \mid \binom{p}{j}$. Also, we have

$$\binom{p+j}{j} = \frac{(p+j)(p+j-1)\cdots(p+1)}{j!} \equiv \frac{(j)(j-1)\cdots(1)}{j!} \equiv 1 \pmod{p}.$$

These imply $\binom{p}{j}\binom{p+j}{j} \equiv \binom{p}{j} \pmod{p^2}$. It follows that

$$\sum_{j=0}^{p} \binom{p}{j}\binom{p+j}{j} \equiv 1 + \binom{2p}{p} + \sum_{j=1}^{p-1}\binom{p}{j} = 1 + \binom{2p}{p} + (2^p - 2) \pmod{p}.$$

It remains to prove that $\binom{2p}{p} \equiv 2 \pmod{p}$. Indeed, by the Vandermonde identity, we have

$$\binom{2p}{p} = \sum_{j=0}^{p} \binom{p}{j}^2 \equiv 1 + 1 = 2 \pmod{p}.$$

This completes the proof. □

Problem 7. Prove that there exist infinitely many composite numbers n such that

$$3^{n-1} \equiv 2^{n-1} \pmod{n}.$$

Solution. We show that the result holds when n is of the form $n = 3^{2^k} - 2^{2^k}$ where $k > 1$. Note that all such numbers are composite as they are divisible by 5.

The result is the same as $3^{2^k} - 2^{2^k} \mid 3^{n-1} - 2^{n-1}$. It suffices to prove $2^k \mid n-1$. Firstly, for $k > 1$, we have $k < 2^k$ by simple induction. Therefore, it remains to prove $2^k \mid 3^{2^k} - 1$. But this holds by the lifting the exponent lemma since

$$v_2(3^{2^k} - 1) = v_2(3^2 - 1) + v_2(2^k) - 1 = 3 + k - 1 = k + 2 > k.$$

This proves the result. □

Problem 8. (CSMO 2022 Grade 11 Problem 7) Prove that for any positive real number λ, there are n positive numbers $a_1, a_2, \ldots, a_n$ ($n \geqslant 2$), so that $a_1 < a_2 < \cdots < a_n < 2^n\lambda$ and for any $k = 1, 2, \ldots, n$ we have

$$\gcd(a_1, a_k) + \gcd(a_2, a_k) + \cdots + \gcd(a_n, a_k) \equiv 0 \pmod{a_k}.$$

Solution. Let c be a sufficiently large positive integer such that $2^{c-2^c-1} < \lambda$. Such a c exists since $\lim_{x\to\infty} 2^{x-2^x-1} = 0$. Let $x_1, x_2, \ldots, x_{2^c}$ be any pairwise coprime odd positive integers. Also, let $n = Mx_1x_2\cdots x_{2^c} + 1$ where M is a sufficiently large integer such that $x_j < 2^n\lambda$ for all j and $n > 2^c + 1$. We claim that the following n integers satisfy all conditions:

$$x_1, x_2, \ldots, x_{2^c}, 2^c, 2^{c+1}, \ldots, 2^{n-2^c+c-1}.$$

Firstly, we must have $x_j < 2^n\lambda$ and $2^{n-2^c+c-1} < 2^n\lambda$ by construction. Secondly, if $a_k = x_j$ for some j, then $(a_i, a_k) = 1$ for all i, and hence

$$\gcd(a_1, a_k) + \gcd(a_2, a_k) + \cdots + \gcd(a_n, a_k) = 1 + 1 + \cdots + 1 = a_k \equiv 0 \pmod{a_k}.$$

If $a_k = 2^j$ for some j, then

$$(a_i, a_k) = \begin{cases} 1 & \text{if } a_i = x_\ell \text{ for some } \ell, \\ 2^\ell & \text{if } a_i = 2^\ell \text{ for some } \ell < j, \\ 2^j & \text{if } a_i = 2^\ell \text{ for some } \ell \geqslant j. \end{cases}$$

Therefore, we have

$$\gcd(a_1, a_k) + \gcd(a_2, a_k) + \cdots + \gcd(a_n, a_k) \equiv 2^c + (2^c + 2^{c+1} + \cdots + 2^{j-1}) + 0 = 2^j \equiv 0 \pmod{a_k}$$

as $a_k = 2^j$. $\square$